



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT2 -TECHNICAL PRESENTATION

MLA0203- FUNDAMENTALS OF MACHINE LEARNING

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Introduction to Decision Theory

What is Decision Theory?

- Decision Theory is a mathematical framework used to **choose the best action when there is uncertainty**.
- In real-world problems, we may not know the exact state of the situation.
- Machine learning models use available data to estimate possible outcomes.
- Decision theory helps convert these probabilities into an **appropriate decision**.

Main Components

1. **States of Nature** – Possible actual outcomes.
2. **Actions** – Decisions that can be taken.
3. **Probability** – Likelihood of each state.
4. **Loss/Cost** – Consequence of making a particular decision.
5. **Decision Rule** – Method used to select the best action.

Example

Email → Spam or Not Spam

Bayes Decision Theory

Bayes Decision Theory provides a method for selecting the decision that has the **minimum expected loss**.

It combines:

- Prior probability
- Observed data
- Posterior probability
- Loss associated with each decision

Bayes' Theorem

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

Where:

- $P(Y|X)$ = Posterior probability
- $P(X|Y)$ = Likelihood
- $P(Y)$ = Prior probability
- $P(X)$ = Evidence

Loss Function and Decision Rules

What is a Loss Function?

A **loss function** measures the penalty or cost associated with making a wrong prediction.

For example, in disease prediction:

Actual	Prediction	Result
Disease	Disease	Correct
Healthy	Healthy	Correct
Healthy	Disease	False Positive
Disease	Healthy	False Negative

Types of Loss

1. 0–1 Loss

- Correct prediction $\rightarrow$ 0 loss
- Wrong prediction $\rightarrow$ 1 loss

For example:

False Positive = Cost 2; False Negative = Cost 10

2. Cost-Sensitive Loss

Different errors have different costs.

Problem: Bayes Decision Theory

Problem

A medical test gives:

$$P(\text{Disease}|X) = 0.20$$

$$P(\text{Healthy}|X) = 0.80$$

Suppose:

- Cost of False Negative = 10
- Cost of False Positive = 2
- Correct decision cost = 0

Step 1: Predict Disease

$$R(\text{Disease}|X) = 2(0.80) = 1.6$$

Step 2: Predict Healthy

$$R(\text{Healthy}|X) = 10(0.20) = 2.0$$

Risk of Disease = 1.6

Risk of Healthy = 2.0

Therefore: Predict $\rightarrow$ DISEASE

Introduction to Information Theory

What is Information Theory?

Information Theory is a mathematical framework for **measuring information and uncertainty** in data.

It was introduced by **Claude Shannon**.

For example:

Tomorrow the sun will rise

→ Very expected

→ Low information

Information Content

$$I(x) = -\log_2 P(x)$$

Information is measured in **bits**.

Entropy

Entropy measures the amount of uncertainty or impurity in a dataset.

$$H(Y) = - \sum_{i=1}^n p_i \log_2(p_i)$$

Where:

- $H(Y)$ = Entropy
- p_i = Probability of class i
- n = Number of classes

Example

Suppose:

Pass = 50%

Fail = 50%

Then:

$$H(Y) = 1 \text{ bit}$$

This represents maximum uncertainty for a binary classification problem.

Information Gain

Information Gain measures how much a feature **reduces uncertainty** about the target variable.

$$IG(Y,X) = H(Y) - H(Y|X)$$

Where:

- $H(Y)$ = Entropy before splitting
- $H(Y|X)$ = Entropy after considering feature X

Decision Tree Application

Decision tree algorithms such as ID3 use information gain to select the best feature for splitting the dataset.

Decision Process

Calculate Parent Entropy ↓ Calculate Entropy after each feature split ↓ Calculate Information Gain ↓

Select feature with highest Information Gain ↓ Create Decision Tree Node

Problem: Calculate Information Gain

Consider 8 students:

Pass = 4

Fail = 4

Therefore:

$$H(Y) = 1$$

Attendance > 75%

Group 1

Pass = 3

Fail = 1

Entropy:

$$H_1 = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$H_1 \approx 0.811$$

Weighted Entropy

$$\begin{aligned} H(Y|X) &= \frac{4}{8} (0.811) + \frac{4}{8} (0.811) \\ &= 0.811 \end{aligned}$$

Group 2

Pass = 1

Fail = 3

$$H_2 \approx 0.811$$

Information Gain

$$IG = 1 - 0.811$$

$$IG = 0.189$$

Conditional Entropy and Mutual Information

Conditional Entropy

Conditional entropy represents the uncertainty remaining in one variable after observing another variable.

$$H(Y|X)$$

Mutual Information

Mutual Information measures the amount of information one variable provides about another.

$$I(X;Y) = H(Y) - H(Y|X)$$

Applications

Mutual information can be used for:

- Feature selection
- Identifying useful variables
- Classification

Applications in ML

Information Theory

- **Decision Trees** → Information Gain
- **Feature Selection** → Select useful features
- **Classification** → Measure uncertainty
- **Anomaly Detection** → Detect unexpected events
- **Data Compression** → Efficient data representation

Problem — Select Best Feature

Given:

$$H(Y) = 0.95$$

Feature A

$$H(Y|A) = 0.40$$

$$IG(A) = 0.95 - 0.40 = 0.55$$

Feature B

$$H(Y|B) = 0.65$$

$$IG(B) = 0.95 - 0.65 = 0.30$$

So, Feature A is selected

Because:

$$0.55 > 0.30$$

Decision Theory vs Information Theory

Decision Theory	Information Theory
Chooses best action	Measures information
Uses probability + cost	Uses probability + entropy
Minimizes risk	Reduces uncertainty
Decision making	Feature selection
Expected Risk	Entropy

Thank
you

